

CAPE COLUMBINE, South Africa

WMO: 687120

Lat: 32.8278S

Lon: 17.8558E

Elev: 55

StdP: 100.67

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	9.1	0.6	4.0	16.9	2.6	4.6	15.4	13.2	12.4	11.5	12.4	4.6	150	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	26.7	16.9	24.8	16.1	22.9	16.1	19.1	22.8	18.4	21.6	17.9	20.6	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	20.0	17.3	12.5	19.4	16.8	12.1	18.9	54.3	23.1	52.4	21.7	50.6	20.7	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	10.5	9.7	DB	6.6	31.6	1.2	1.3	5.8	32.5	5.1	33.3	4.4	34.0	3.5	34.9	(4)
			WB	5.0	20.3	1.2	0.7	4.1	20.8	3.5	21.2	2.8	21.6	2.0	22.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	17.4	17.6	16.9	16.4	14.8	13.8	13.3	13.2	14.3	15.3	15.9	17.1
	DBStd	2.50	1.46	1.71	2.13	3.00	2.03	1.97	2.12	1.79	2.09	1.75	1.53	1.66
	HDD10.0	1	0	0	0	0	1	0	1	0	0	0	0	0
	HDD18.3	1113	37	32	56	78	112	139	157	159	125	98	74	46
	CDD10.0	2003	229	212	214	191	151	113	102	101	129	163	178	219
	CDD18.3	73	8	11	12	18	3	2	1	1	4	3	2	7
	CDH23.3	339	16	40	72	101	14	6	2	6	33	17	10	23
	CDH26.7	66	2	5	16	22	1	0	0	0	10	4	2	3
Wind	WSAvg	5.4	6.2	5.8	5.4	5.1	4.6	4.8	4.8	5.0	5.4	5.7	6.3	6.1
Precipitation	PrecAvg	259	6	6	7	20	32	50	45	39	24	11	10	7
	PrecMax	377	37	26	26	71	85	175	111	83	77	49	41	24
	PrecMin	104	0	0	0	3	3	7	7	9	4	0	1	0
	PrecStd	61	8	8	7	17	20	32	23	19	19	12	11	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	28.4	29.0	29.5	25.5	23.8	22.6	23.6	27.5	26.3	24.2	26.2
		MCWB	17.8	18.6	18.7	17.7	14.3	14.3	13.3	14.5	14.9	15.9	15.9	17.0
	2%	DB	22.4	24.7	25.2	26.0	21.9	20.4	20.2	19.3	21.5	22.2	21.4	22.6
		MCWB	18.0	17.3	17.4	16.0	13.6	13.1	12.9	12.4	13.2	14.5	15.5	17.4
	5%	DB	21.1	22.2	22.1	23.6	19.9	18.0	18.5	17.2	18.2	19.4	19.7	21.1
		MCWB	17.5	17.6	17.2	15.4	13.4	12.8	12.3	12.3	13.3	14.6	15.5	17.2
	10%	DB	20.1	20.8	20.0	20.8	17.7	16.2	16.6	15.7	16.7	18.0	18.7	20.1
		MCWB	17.0	17.3	16.8	14.7	13.5	12.7	12.3	12.8	13.5	14.5	15.4	16.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.9	19.7	18.7	16.4	15.6	15.2	15.5	16.0	17.1	17.7	19.2
		MCDB	22.8	23.8	25.1	27.0	19.7	18.8	18.2	18.7	21.7	22.7	20.7	22.5
	2%	WB	18.7	19.0	18.6	17.4	15.6	14.9	14.4	14.5	15.0	16.1	16.9	18.4
		MCDB	21.2	22.6	22.5	22.9	18.0	17.2	16.5	16.4	17.9	19.5	19.5	21.3
	5%	WB	18.1	18.2	17.8	16.6	15.1	14.3	13.8	13.8	14.5	15.5	16.3	17.7
		MCDB	20.2	20.9	20.9	20.6	16.9	16.0	15.8	15.5	16.8	18.1	18.8	20.3
	10%	WB	17.5	17.6	17.2	15.9	14.6	13.7	13.3	13.3	14.0	14.9	15.7	17.2
		MCDB	19.5	19.9	19.6	19.0	16.3	15.3	15.1	14.8	15.9	17.1	18.1	19.4
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.6	5.2	5.2	4.5	4.6	4.9	4.8	5.1	5.3	5.4	5.6
		MCDBR	6.8	8.1	8.1	9.6	8.9	7.4	8.4	8.0	7.9	7.4	6.8	6.9
		MCWBR	3.4	3.7	3.3	3.7	3.6	4.1	4.5	4.4	3.9	3.6	3.4	3.4
	5% WB	MCDBR	5.9	6.6	6.4	7.1	5.3	4.5	5.3	5.2	5.6	5.8	5.9	6.0
MCWBR		3.4	3.6	3.4	3.5	3.3	3.1	3.6	3.5	3.5	3.3	3.4	3.4	
Clear-Sky Solar Irradiance	taub	0.335	0.336	0.329	0.317	0.315	0.308	0.311	0.327	0.336	0.346	0.331	0.335	
	taud	2.564	2.571	2.584	2.610	2.607	2.592	2.581	2.539	2.472	2.463	2.551	2.554	
	Ebn at Noon	1003	983	954	913	863	842	857	888	931	961	1001	1006	
	Edh at Noon	107	103	95	83	74	70	75	87	104	113	108	109	
All-Sky Solar Radiation	RadAvg	8.33	7.48	6.07	4.46	3.13	2.70	3.00	3.75	5.12	6.61	7.82	8.40	
	RadStd	0.23	0.21	0.21	0.23	0.21	0.15	0.17	0.16	0.19	0.28	0.22	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61

Nomenclature: See separate page